

SHAO-CHUN HUNG

Computer Vision | Machine Learning | Applied AI

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Summary

Graduate researcher in Mathematical Modeling and Scientific Computing at NYCU, with hands-on experience in machine learning, computer vision, data analysis, and AI workflow prototyping. Built end-to-end project pipelines spanning data preprocessing, modeling, evaluation, and experimentation, from YOLO-based ship detection to graph-based MRT passenger flow analysis.

Skills

Programming Python · Matlab · Git · SQL	Data Analysis Data Visualization · Pandas · Numpy
AI / Modeling ML · PyTorch · Computer Vision · YOLO · OpenCV	Tools GitHub · Docker · API Integration
LLM / Workflow Harness Engineering · AI-Agents-Assisted Workflow	

Research & Projects

Optimizing Ship Detection in Multi-Scale Environments	NYCU
- Developed a ship detection pipeline for multi-scale environments and evaluated model performance across complex visual scenes. - Improved the YOLO11 architecture by comparing feature extraction modules by using Python and PyTorch. - Built a GUI tool to support image enhancement, annotation, copy-paste augmentation, and post-training prediction analysis for the detection workflow. - Achieved a +1.3 mAP@50 improvement over baseline and provided structured analysis across vessel categories, inference speed, and detection accuracy.	03/2025 - Present
Passenger Flow Analysis of the Taipei MRT	NTNU
- Built a passenger flow analysis project for the Taipei MRT using open government transportation data. - Constructed a 119 × 119 station transfer matrix and applied Markov Chain modeling, PageRank, and Power Method iteration. - Identified high-mobility interchange hubs and surfaced regional transit patterns through algorithmic analysis.	10/2022 - 07/2023
CPBL Batting and Pitching Performance Analytics	NTNU
- Built a structured analytics project for CPBL batting and pitching performance using multi-season official statistics. - Parsed and cleaned raw web data, then designed a cross-year comparison pipeline with conditional filtering rules in Python. - Identified recurring performance patterns across three consecutive championship-winning teams and delivered end-to-end analytical reporting.	10/2021 - 01/2022

Awards

1st Place — 2026 TSMC IT CareerHack
- Worked on legacy code refactoring and helped establish a more verifiable and reproducible refactoring process. - Designed and optimized spec-driven prompts to improve task alignment and output structure. - Converts C to Rust and migrates MVC to decoupled architecture, reducing refactoring cycle time by ~80%.
Best Research Award — 2022 Applied Mathematics Conference
- Led project execution : mathematical model derivation, Python implementation, and results interpretation. - Helped address trip-planning challenges for tourists by revealing major interchange hubs and passenger flow patterns

Experience

Teaching Asssistant, Calculus NYCU	09/2024 - Present
- Developed an AI-agent-assisted workflow to support student registration and improve class administration efficiency. Awarded Best Teaching Assistant of Academic Year 2025	

Education

National Yang Ming Chiao Tung University (NYCU)	Hsinchu, Taiwan
M.S. — Mathematical Modeling and Scientific Computing	09/2024 - Present
National Taiwan Normal University (NTNU)	Taipei, Taiwan
B.S. — Mathematics	09/2020 - 06/2024
GPA : 4.0/4.3 Department Rank : Top 5	